CS5013 Project Proposal

Hostel Complaint & Attender Duty Tracking System

Ajoy Mathew, [CS23B101]

September 6, 2026

1 Team information

Team Structure: Solo

- **Name:** Ajoy Mathew
- **Roll No.:** CS23B101
- **Department:** Computer Science and Engineering
- **Primary Contact:** cs23b101@smail.iitm.ac.in
- **GitHub:** ajoymathew07

2 Problem statement

Hostel managers at IIT Madras manually process daily maintenance complaints submitted by residents via Google Forms by copying entries into spreadsheets and sorting them into domains (Electrical, Civil, RO Plant). The manager must manually verify attender availability, cross-reference floor allocations across attenders, track inspection IDs, and tally resolution statuses. Furthermore, scoring attender performance (+2 points for resolution, custom points for special duties, and -1 point for infractions like late arrival or forgotten IDs) for monthly bonus calculations is entirely manual, error-prone, and consumes 30–45 minutes of administrative time every day.

3 Users and stakeholder

Named stakeholder: Hostel Manager, Cauvery Hostel, IIT Madras.

I met the Cauvery Hostel Manager at the hostel office to understand the daily complaint-handling workflow. The manager explained that managing four attenders across multiple floors involves tedious manual spreadsheet reconciliation, constant verbal check-ins for inspection status, and manual calculation of bonus points and penalty deductions at the end of the month. The manager had to copy complaints from a google form response, categorize them, assign it to a attender and track the redressal which takes daily 30-45min. The manager expressed a strong need for an automated ingestion and duty-tracking tool that automates categorization and floor-based assignment while providing a simple interface for attenders to update ticket statuses and check their standings.

The stakeholder's acknowledgement email is attached as Appendix A.

4 Prior work and why it does not suffice

1. **IITM CMS EU App (Campus Management System – Engineering Unit)** – *Institute-level central portal.*
   Overlap: Central app where campus residents and departments raise official infrastructure complaints and obtain a formal tracking/docket number.
   Gap & Distinction in Functionality: The IITM CMS EU App is designed for high-level campus-wide administrative tracking and contract dispatch by the engineering unit. It does not address the internal operational logistics of an individual hostel. Our system stores the CMS tracking number alongside each internal complaint, bridging institutional audit compliance with local hostel-level task execution.

2. **Manual Google Sheets Workflow** – *Internal / manual incumbent.*
   Overlap: Records complaints in tabular format, tracks resolution notes, and sums basic points using spreadsheet formulas.
   Gap: Requires continuous manual copy-pasting from Google Form responses; lacks automated floor-load balancing and real-time attender availability checks; easily corrupted when multiple staff view or edit formulas from mobile devices.

3. **Commercial Helpdesk Platforms (e.g., Freshdesk / Zoho Desk)** – https://freshdesk.com.
   Overlap: Provides automated ticketing, category routing, and agent assignment.
   Gap: High recurring subscription costs; over-engineered for hostel maintenance; requires individual resident logins; lacks custom floor-to-attender dynamic duty mapping, penalty deductions, and the specific bonus gamification system required by the hostel.

4. **Open-Source Issue Trackers (e.g., Redmine / Request Tracker)** – https://www.redmine.org.
   Overlap: Self-hostable ticket lifecycle tracking with customizable fields and status flows.
   Gap: Requires heavy server setup and maintenance; lacks native integration with Google Sheets for non-technical office staff; lacks mobile-optimized duty cards for attenders and built-in automated performance-bonus calculations.

5 System Scope & Technical Architecture

To ensure the project provides substantial systems-level and algorithmic engineering rigor beyond a standard CRUD application, the system implements the following architecture:

1. **Automated Complaint Ingestion, Prioritization & Categorization:** The web application synchronizes directly with the Google Form response sheet, automatically ingesting new complaints, assigning baseline priorities, and adding them into relevant domains (Electrical, Civil, RO Plant, Network) without manual copy-pasting.
2. **Heuristic Attender Dispatch & Dedicated Task Views:** The system executes a routing heuristic factoring in physical floor distribution, real-time attender availability, and active pending workloads to assign each complaint to the most appropriate attender. The allocated task immediately populates on that attender's dedicated mobile interface.
3. **Field Verification :** The attender inspects the reported issue on-site, verifies the defect, and logs the complaint on the official IITM EU CMS App. The attender then inputs the received CMS tracking docket number and current status directly into the web app, establishing end-to-end institutional traceability.
4. **Resolution Workflow, Leaderboard & Custom Duties:** Once the repair is completed, the attender marks the task as resolved on the web app, automatically earning resolution credits (+2 points) that reflect on the dynamic monthly leaderboard. Additionally, the man-

ager can post ad-hoc special duties with custom point values (e.g., hostel night preparation, meeting refreshments) open to any attender.

5. **Penalty Management, Audit Logging & Real-Time Notifications:** A dedicated penalty interface enables the manager to log infractions (e.g., unpunctuality, incomplete tasks, unverified absences) with point deductions (-1). All state changes, point allocations, and penalty adjustments are recorded in an append-only audit log, with an administrative notification panel showing real-time duty updates.
6. **Manager Dashboard & Excel Export:** The manager has access to a centralized dashboard providing comprehensive daily summaries of tasks completed per attender, active backlogs, and throughput trends. The manager can generate and export daily and weekly reports of registered and resolved complaints directly into formatted Excel spreadsheets (`.xlsx`) for official hostel office documentation.

In addition to the core pipeline, the system provides dedicated administrative controls, collaborative task support, and telemetry intelligence:

- **Manager Oversight Dashboard & Audit Export:** Provides real-time summaries of daily tasks executed by each attender, daily/weekly resolution throughput, and one-click dynamic multi-tab Excel export (`.xlsx` via Apache POI) for hostel office records.
- **Shared-Task Concurrency Control:** Supports collaborative maintenance tasks (e.g., common room overhauls, furniture transport) allocated to multiple attenders concurrently. Enforces strict transactional isolation and optimistic locking to prevent race conditions and double-crediting when multiple attenders update shared sub-tasks across intermittent mobile connections.
- **Log Intelligence & Telemetry Analytics:**
  - *Bottleneck Detection:* Aggregates timestamp intervals across ticket lifecycle states to compute Mean Time to Resolution (MTTR) and identify bottlenecks of each category of complaints.
  - *Operational Anomaly Detection:* Flags statistical outliers, including suspiciously prolonged resolution latencies, repeated complaints originating from identical locations (chronic fixture defects / band-aid repairs), and sudden volume surges across wings.
  - *Administrative Visualizations:* Displays rolling trends for arrival velocities, active attender workloads, SLA breach distributions, and anomaly alerts.

6 Tech stack

Language: **Java 17 / 21**.

- **Build system:** Maven (`pom.xml`)
- **Framework:** Spring Boot 3.x (Spring MVC, Spring Data JPA, Embedded Tomcat)
- **Key libraries & dependencies:**
  - Google Sheets API Client (`google-api-services-sheets:v4`) for automated delta ingestion
  - Apache POI (`poi-ooxml`) for dynamic multi-tab Excel report generation
  - SQLite / Embedded H2 (local development) & PostgreSQL Driver (cloud production persistence)
  - Spring Task Scheduling (`@Scheduled`, `@Async`) for background polling, SLA aging, and telemetry scans
  - Spring Validation (`hibernate-validator`) for entity constraints, CMS ID format checks, and OCC versioning
  - Spring WebSocket / SSE for real-time manager notification broadcasts
  - Apache OpenNLP for rule-assisted text categorization and entity extraction

  - Thymeleaf + Tailwind CSS with Chart.js for responsive desktop/mobile interfaces and analytical dashboards
- **Where the tool will run:** Standalone executable `.jar` hosted on a lightweight cloud PaaS container (Render/Railway) for mobile access by attenders, with a local-server fallback (`http://localhost:8080`) executable directly on the manager's desktop incase of an outage.

7 AI-usage statement

I will use AI coding assistants (such as Claude Code, GitHub Copilot, ChatGPT, or Gemini) during this project, in line with the course policy on AI-assisted programming. Every line of code I submit will be code that I understand and can explain in the viva, whether I typed it or the assistant generated it.

8 Verification plan

Test fixtures we will use:

- A sample of real complaint logs from Cauvery Hostel with student identities and personal credentials anonymized with dummy data.
- A synthetic corner-case test suite covering edge cases: ambiguous complaint categories, identical simultaneous submissions, unallocated floor numbers, and scenarios where all assigned floor attenders are marked unavailable.
- Performance fixture simulating concurrent mobile status updates from 4 attenders simultaneously resolving tickets.

User-visible acceptance criteria the stakeholder will use to say "yes, this works":

- Ingests new raw Google Form responses automatically via API polling without manual copy-pasting or duplicate processing.
- Automatically categorizes complaints into Civil, Electrical, and RO Plant with at least 90% accuracy and assigns them to the primary floor attender.
- Attenders can open their assigned queue on mobile, enter an Inspection ID, and mark issues resolved, correctly crediting +2 points.
- Manager can log penalties (-1) and special duties, view the real-time leaderboard, and export a clean `.xlsx` report with one click.
- Comprehensive Manager dashboard with working anamoly detection, bottleneck identification and visualizations.

9 Milestone plan

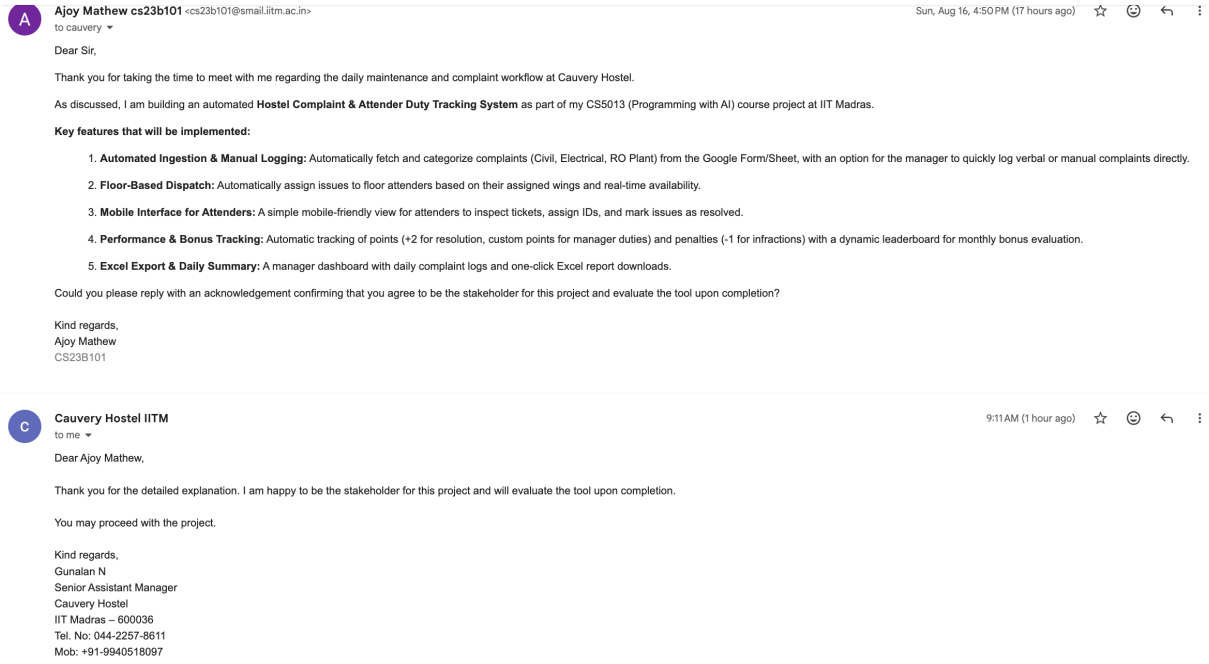
- **By the design doc (11 Sep 2026):** System architecture and module breakdown (Ingestion, Dispatch, Concurrency/Ledger, Analytics); relational schema with optimistic locking and audit tables; formal state machine for complaints and CMS docket tracking; dispatch heuristic formulation; frontend layout and data contract specifications (manager desktop vs. attender mobile); test plan covering synthetic fixtures and concurrency stress cases; and a revised timeline.
- **By the mid-demo (9 Oct 2026):** Working core pipeline: automated Google Sheets ingestion with offline CSV fallback; rule-based categorization; dynamic priority aging and floor-dispatch heuristic; attender mobile interface with CMS ID logging and +2 point attribution; functional manager dashboard displaying real-time attender task lists; and initial cloud deployment (`.jar` on Render/Railway).

- **By the final submission (6 Nov 2026):** Full operational system: manager special duty assignments, penalty deduction module (−1) with audit logs, shared-task concurrency control, telemetry analytics (bottlenecks, MTTR, anomaly alerts with Chart.js), monthly leaderboard, one-click Excel export (`.xlsx`), test verification, and stakeholder sign-off.

10 Risks and plan B

1. **Risk:** Google Sheets API quota limits, authentication expiry, or network downtime on the office desktop.
   Plan B: Provide an offline manual CSV drag-and-drop ingestion interface in the manager dashboard as a backup sync mechanism.
2. **Risk:** Student complaint text is too ambiguous or misspelled for auto-categorization rules.
   Plan B: Implement an 'OTHER / UNASSIGNED' triage fallback queue allowing the manager to set the category and assign the attender with a single click.
3. **Risk:** Attenders encounter network issues or friction using the mobile web view while inspecting hostel wings.
   Plan B: Include an "Assisted Admin Log" on the manager's desktop interface so the manager can update inspection IDs and resolutions during daily roll-call check-ins.

A Stakeholder acknowledgement email



Ajoy Mathew cs23b101 <cs23b101@smail.iitm.ac.in> Sun, Aug 16, 4:50 PM (17 hours ago)
to cauvery

Dear Sir,

Thank you for taking the time to meet with me regarding the daily maintenance and complaint workflow at Cauvery Hostel.

As discussed, I am building an automated **Hostel Complaint & Attender Duty Tracking System** as part of my CS5013 (Programming with AI) course project at IIT Madras.

Key features that will be implemented:

1. **Automated Ingestion & Manual Logging:** Automatically fetch and categorize complaints (Civil, Electrical, RO Plant) from the Google Form/Sheet, with an option for the manager to quickly log verbal or manual complaints directly.
2. **Floor-Based Dispatch:** Automatically assign issues to floor attenders based on their assigned wings and real-time availability.
3. **Mobile Interface for Attenders:** A simple mobile-friendly view for attenders to inspect tickets, assign IDs, and mark issues as resolved.
4. **Performance & Bonus Tracking:** Automatic tracking of points (+2 for resolution, custom points for manager duties) and penalties (-1 for infractions) with a dynamic leaderboard for monthly bonus evaluation.
5. **Excel Export & Daily Summary:** A manager dashboard with daily complaint logs and one-click Excel report downloads.

Could you please reply with an acknowledgement confirming that you agree to be the stakeholder for this project and evaluate the tool upon completion?

Kind regards,
Ajoy Mathew
CS23B101

Cauvery Hostel IITM 9:11 AM (1 hour ago)
to me

Dear Ajoy Mathew,

Thank you for the detailed explanation. I am happy to be the stakeholder for this project and will evaluate the tool upon completion.

You may proceed with the project.

Kind regards,
Gunalan N
Senior Assistant Manager
Cauvery Hostel
IIT Madras – 600036
Tel. No: 044-2257-8611
Mob: +91-9940518097

Figure 1: Project acknowledgement by stakeholder